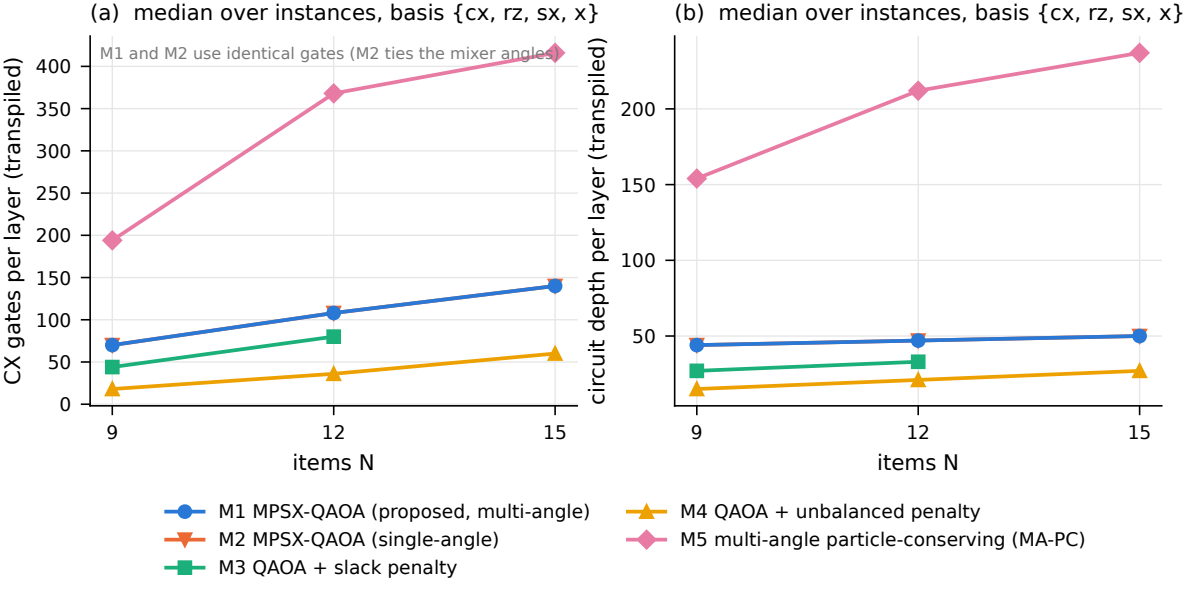
(a) median over instances, basis {cx, rz, sx, x}
M1 and M2 use identical gates (M2 ties the mixer angles)
CX gates per layer (transpiled)
0
50
100
150
200
250
300
350
400
9
12
15
items N
(b) median over instances, basis {cx, rz, sx, x}
circuit depth per layer (transpiled)
50
100
150
200
9
12
15
items N
M1 MPSX-QAOA (proposed, multi-angle)
M2 MPSX-QAOA (single-angle)
M3 QAOA + slack penalty
M4 QAOA + unbalanced penalty
M5 multi-angle particle-conserving (MA-PC)